



CausalBridgeQA: a causal inference-based approach for robust enhancement of multi-hop question answering

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Abstract

Multi-Hop Question Answering (MHQA) tasks require retrieving and reasoning over multiple relevant supporting facts to answer a question. However, existing MHQA models often rely on a single entity or fact to provide an answer, rather than performing true multi-hop reasoning. Additionally, during the reasoning process, models may be influenced by multiple irrelevant factors, leading to broken reasoning chains and even incorrect answers. In recent years, causal inference-based methods have gained widespread attention in bias removal research. But existing models still perform poorly when dealing the complex causal biases hidden in multi-hop evidence. To address these challenge, we propose CausalBridgeQA, a novel method that integrates multi-hop question answering with causal relationships, effectively mitigating feature spurious correlations and the problem of broken reasoning chains. Specifically, we first extract causal relationships from the input text context, then compile these relationships into causal questions containing higher-level semantic information and feed them into MHQA reasoning system. Finally, we design a knowledge compensation mechanism in the reading comprehension module of the MHQA system to specifically address questions that are difficult to answer or frequently answered incorrectly, significantly improving the performance of MHQA tasks. Finally, a series of experiments conducted on three real-world QA datasets verified the effectiveness of our proposed method.

Keywords

multi-hop question answering; causal inferenc; explainable artificial intelligences

1 Introduction

Multi-Hop Question Answering (MHQA) reasoning is one of the most challenging tasks in the field of Natural Language Processing (NLP). It requires the model to retrieve and reason over multiple relevant pieces of supporting information from large knowledge bases or documents to answer a question. Unlike traditional question answering tasks, which typically rely on a single fact or paragraph to provide a direct answer [1,2], MHQA requires model to find candidate answers across multiple different relevant contextual paragraphs and reason over these related facts to arrive at the correct answer [3,4].

Currently, the main challenges faced by the MHQA reasoning task focus on the following aspects. First, when retrieving relevant supporting facts, many pieces of evidence come from distant facts within the document, which are interrelated and need to be effectively integrated to form a correct and complete candidate answers [5]. Second, during the reasoning process, the model is

prone to being interfered with by spurious correlations in features, leading to reasoning errors. When searching for document contexts that match the entities in the question, the complex entity relationship network in the document may cause the model to get stuck and produce incorrect answers, even leading to model degradation, where the model always provides an answer based on a single supporting fact [6]. Third, in multi-hop reasoning, the model needs to maintain the integrity of the reasoning chain across different reasoning steps, meaning each piece of evidence should build upon the previous one. The model must handle challenges such as fact ambiguity, irrelevant information, and errors in cross-hop reasoning. In summary, the MHQA reasoning task requires building a sufficiently powerful and complex neural network that fully understands and models contextual semantic information to address these challenges [7].

To address the aforementioned issues, existing research work primarily focuses on the following aspects: Firstly, represented by methods that introduce complex reasoning modules [8–10], these

methods aim to improve task performance by enhancing retrieval mechanisms and integrating pre-trained language models. Although these methods partially address the challenge of retrieving distant facts, but they still struggle to understand complex entity relationships. Secondly, methods represented by graph neural networks [11–13], these methods explicitly model the relationships between supporting facts, enhancing the model’s performance in cross-hop reasoning. However, they still face a high risk of errors when handling ambiguous facts or misleading information. Finally, methods represented by causal inference [14,15] and interpretability [16,17] aim to explore the reasoning chains and transparency of the model’s reasoning process, without placing much emphasis on task performance. Moreover, in methods represented by causal inference [18–20], most efforts focus on enhancing training datasets through additional annotations or adversarial examples, including tasks like counterfactual generation [21]. Researchers also tend to focus on causal relationship identification [22,23] in event datasets, with relatively few studies on QA datasets. This is because extracting reasonable causal relationships from QA datasets is not straightforward [24].

Although MHQA has been extensively studied, existing methods overlook a critical fact: many questions in QA datasets inherently explore causal relationships between two entities. However, due to the difficulty of accurately identifying these causal relationships, current multi-hop QA models struggle in reasoning processes, hitting performance bottlenecks. Based on this, we argue that extracting implicit causal relationships from text, or transforming questions into a form enriched with higher-level “causal semantic information” and using these extracted causal relationships as supplementary knowledge for the model, can significantly enhance the performance of MHQA tasks.

As shown in Fig. 1, the upper part illustrates the MHQA task, while the lower part demonstrates causal extraction. We extract causal relationship pairs from the background information in “Context 2” to strengthen reasoning chains in a more structured manner. Additionally, by iteratively integrating all extracted causal pairs, the final question is reformulated into a “causal-information-enriched question form” (e.g., “What events led to the French ace pilot and adventurer, who flew the L’Oiseau Blanc, attempting the transatlantic flight to win the Orteig Prize?”), which asks about causal relationship. Our approach further uncovers the implicit causal links in the text, more precisely capturing the reasoning chains, thereby improving the model’s ability to accurately address complex questions. In summary, we make the following contributions:

- We propose a novel approach named CausalBridge-QA to address the issues of reasoning chain breaks and feature spurious correlations in MHQA task. To the best of our knowledge, we are the first to explore context relationships from a causal inference perspective and apply them to MHQA tasks.
- We propose a “knowledge compensation” mechanism specifically designed to address complex questions that can easily lead the model to make errors or struggle.

Question: Which French ace pilot and adventurer fly L’Oiseau Blanc? Context 1: Levasseur PL.8_0 Levasseur built the aircraft L’Oiseau Blanc in 1927, specifically for pilots Charles Nungesser and Francois Coli for a transatlantic attempt to win the Orteig Prize. Context 2: Charles Nungesser_0 Charles Eugène Jules Marie Nungesser (15 March 1892 – presumably on or after 8 May 1927) was a French ace pilot and adventurer, best remembered as a rival of Charles Lindbergh. Context 3: L’Oiseau Blanc_0 L’Oiseau Blanc was a French biplane flown by Charles Nungesser and Francois Coli in 1927 during their attempt at the first non-stop transatlantic flight from Paris to New York. Answer: Charles Eugène Supporting Facts: [“Levasseur PL.8_0”], [“Charles Nungesser_0”], [“L’Oiseau Blanc_0”]
Context Passage2: Levasseur built the aircraft L’Oiseau Blanc in 1927, specifically for pilots Charles Nungesser and Francois Coli for a transatlantic attempt to win the Orteig Prize. Extracted Cause: The transatlantic attempt by Charles Nungesser and Francois Coli to win the Orteig Prize. Extracted Effect: Levasseur designed and built aircraft L’Oiseau Blanc. Final Extracted Cause: Charles Nungesser and Francois Coli decided to attempt a transatlantic flight to win the Orteig Prize. Final Extracted Effect: Levasseur designed and built the aircraft L’Oiseau Blanc to support their attempt. Final Generated Question: What events led to the French ace pilot and adventurer, who flew the L’Oiseau Blanc, attempting the transatlantic flight to win the Orteig Prize?

Fig. 1 An example from the HotpotQA dataset in the multi-hop QA task. Brown and pink indicate the key entity information of the question, green represents important information in causal reasoning, and orange denotes miscellaneous distracting entities

- We conduct a series of experiments on three real and effective datasets: HotpotQA, SQuAD, and TQA. The experimental results demonstrate the effectiveness and superiority of our proposed method.

2 Related work

In this section, we primarily introduce the representative research works in the field of MHQA reasoning tasks and categorize them into the following three types based on their research methods for discussion.

• Pre-trained model-based methods

Nishida et al. first proposed QFE [25], which uses an RNN with an attention mechanism on the question to sequentially extract evidence sentences. Subsequently, Zhao et al. proposed Transformer-XH [8], which uses eXtra Hop attention to enable intrinsic modeling of structured texts in a fully data-driven way. Meanwhile, Tu et al. proposed SAE [6], SAE filters distracting information using a document classifier trained with pairwise learning-to-rank loss, then feeds the selected answer-relevant documents into model to jointly predict answers and supporting sentences. Recently, Li et al. proposed FE2H [26], which divides the document selector and reader into two stages in an easy-to-difficult manner. By integrating large-scale pre-trained models, it achieves state-of-the-art performance in MHQA tasks.

• Graph neural network-based methods

DFGN [5] was first proposed by Qiu et al., which is a Dynamic Fusion Graph Network that starts from entities mentioned in the given query, explores along an entity graph dynamically constructed from the text, and gradually identifies relevant supporting entities from the given document. Subsequently, Fang et al. introduced HGN [11], a Hierarchical Graph Network capable of aggregating different granularity levels (question, paragraph, sentence, entity) and performing multi-hop reasoning by updating node representations through graph propagation. Both AMGN [27] and C2F Reader [12] improve MHQA performance through self-attention mechanisms within the graph.

• NLP model methods interpretability and causal inference

Due to the multi-hop reasoning characteristics of MHQA tasks being highly aligned with the need for transparent decision-making in practical applications, this task has also become an ideal testing ground for studying model interpretability, attracting significant attention from researchers. Yang et al. [4] first proposed the HotpotQA dataset, an explainable dataset specifically designed for multi-hop QA scenarios, which has been widely used in numerous research studies. Kundu et al. proposed a path-based reasoning method called PathNet [28], which can explicitly find the reasoning paths of the model and score them, without the need for any supervision. Path-GCN [29] combines reasoning chains and entities to build a graph and then uses Gated-RGCN to reason over it. Recently, [18] and [19] have respectively improved the performance of MHQA tasks and enhanced their interpretability from the perspective of causal inference, utilizing counterfactual generation and debiasing methods in QA.

■ 3 Method

In this section, we first introduce the overall framework of the proposed CausalBridgeQA, followed by a detailed explanation of its two main processing stages: Causal Relationship Extraction (CRE) and Multi-Hop Question Answering Reasoning (MHQA). We will elaborate on the methods used in each stage and the interaction

between their internal components, concluding with an explanation of how these methods and components are effectively applied to MHQA tasks to address real-world problems.

3.1 Overall model architecture

The overall framework of CausalBridgeQA is illustrated in Fig. 2. On the left is the input document (with an example from HotpotQA), and on the right, we illustrate the two main steps of the framework, CRE and MHQA.

In the CRE stage, we first preprocess the input data and use external knowledge bases (such as CauseNet) and logical rules to guide the extraction of causal relationships. The original input is processed using an improved rule extraction method, where entity relationships are converted into causal graphs with pointers to capture both intra- and inter-sentence causal dependencies. Then, we use a generation model fine-tuned on HotpotQA to generate questions Q that inquire about causal relationships. These questions Q have a semantic structure similar to the original input q , but they specifically ask about causal relationships. Finally, we use both the question Q and the extracted causal relationships as input to the MHQA module's “knowledge compensation” to address those challenging questions.

In the MHQA phase, we first filter out irrelevant information from the documents. Specifically, we use the “[CLS]” token output from BERT [30] as the vector representation for each question/document pair. For particularly large contexts, we also employ TF-IDF to assist in retrieving and initializing candidate paragraphs and calculating relevance scores for ranking documents, ultimately identifying the golden documents. In reading comprehension module, we initially obtain the predicted answers following the standard process. For questions that are answered incorrectly or left unanswered, we incorporate the causal relationships extracted during the CRE phase along with the question Q as supplementary information, reintroducing them into the training process iteratively. We design a “rollback” mechanism to refine the process and dynamically integrate the causal relationships learned during the CRE phase. As a result, we successfully infuse causal knowledge and accurately answer complex questions.

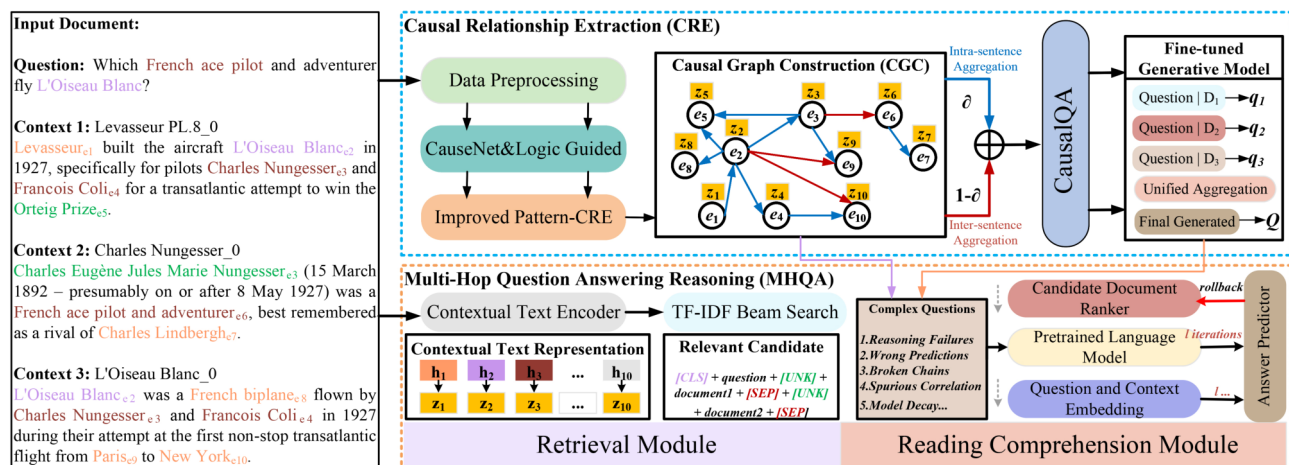


Fig. 2 Overall CausalBridgeQA framework. The input document first goes through CRE to extract causal knowledge and reformulate the question, then in the MHQA phase, it supplements the training process to improve performance

3.2 Causal relationship extraction (CRE)

3.2.1 Data preprocessing

In the HotpotQA dataset, answers are classified into three types: yes, no, and text span. We remove yes/no type questions and contexts as they are irrelevant to our research. Additionally, following the suggestions of [14] and [31], we adopt a data-driven approach at a practical level to filter meaningful questions based on causal question categories and semantic dimensions. For example, the typology contains eight main categories, based on the causal link which is utilized to express the causal relationship, e.g., Adverbial (“consequently”, “accordingly”, “therefore”), Prepositional (“because of”, “due to”), Subordination (“because”, “as”, “since”), and Clause-integrated linkage (“that’s why”, “the result was”). After examining the 360 causal relationships extracted from HotpotQA, we effectively modify them using CauseNet [15] in combination with logical rules, reducing the number of relationship types extracted from HotpotQA to 280.

3.2.2 Pattern based causal-effect link extraction

Let $S = \{s_1, s_2, \dots, s_n\}$ be the set of all sentences in the context. Each sentence s_i is composed of a sequence of words $w_1, w_2, \dots, w_m$, that is $s_i = [w_1, w_2, \dots, w_m]$.

Let the set of patterns be $P = \{p_1, p_2, \dots, p_k\}$, and each pattern p_j is defined as a tuple $p_j = (T_j, C_j)$, where T_j is the core vocabulary set of the pattern, e.g., $T_j = \{\text{“because”, “due to”}\}$, C_j is the set of constraint rules for the pattern. The set of constraint rules $C_j = \{c_1, c_2, \dots, c_l\}$, where each constraint c_k describes the grammatical, semantic, or positional restrictions of words or phrases in the sentence $c_k = (\text{Position}, \text{ConstraintType})$. Position indicates the location of the constraint, such as “left of causal words” or “end of the sentence.” ConstraintType is defined as a set of conditions, such as POS ($w \in \{\text{noun, verb, adjective}\}$). Sem ($w \in \{\text{cause, effect}\}$). Syntax ($w = \text{NP (noun phrase)}$). We define the matching function between sentence s and pattern p_j as:

$$\text{Match}(s, p_j) = \begin{cases} 1, & \text{if } s \text{ contains } T_j \text{ and satisfies } C_j; \\ 0, & \text{otherwise.} \end{cases}$$

For each sentence $s_i \in S$, we calculate the matching value of each pattern p_j in the pattern set P .

$$M(s_i) = \{p_j \mid \text{Match}(s_i, p_j) = 1, \forall p_j \in P\}.$$

For the matched pattern p_j , extract the “Cause” part and the “Effect” part of the causal relationship. Define the extraction function as:

$$\text{Extract}(s_i, p_j) = (\text{Cause}(s_i, p_j), \text{Effect}(s_i, p_j)).$$

In CRE module, e_1 represents the “cause” in the causal relationship, i.e., the factor that leads to the occurrence of an event. z_1 represents the “effect” in the causal relationship.

Finally, the causal relationship set R is defined as:

$$R = \bigcup_{s_i \in S} \bigcup_{p_j \in M(s_i)} \text{Extract}(s_i, p_j).$$

3.2.3 Cause and effect question generation

After extracting causal relationships from the input text, we feed them, along with the initial question q , into a pre-trained generative model and use the text generated by the Effect as the answer. Specifically, we utilize ProphetNet [32], a model designed for “Predicting Future N-gram for Sequence-to-Sequence Pre-training.” The advantage of this approach is that its ability to generate text by predicting the next n-gram instead of just the next token, which helps to prevent overfitting on strong local correlations. In addition, given the excellent performance of large models, we choose ChatGPT-4o [33] to help us process the HotpotQA data. We select the top 100 test samples from the HotpotQA dataset and convert the JSON-format data to CSV, making it easier for us to quickly evaluate the quality of the model’s generated results.

As shown in CRE in Fig. 2, we first identify the initial question q , and then perform causal extraction on each context related to the question. The extracted causal relationships are output according to the CausalQA method (one corresponding to the cause and the other to the effect). Therefore, each context generates two questions, cause question and effect question. Here, we continuously aggregate the semantic information of these two questions, and treat the initial question as the main objective, ultimately obtaining a question Q containing rich semantic information. Q structurally incorporates both the initial question and the causal relationships associated with the entities in the initial question.

3.3 Multi-hop question answering reasoning (MHQA)

3.3.1 Retrieval module

The primary function of the retrieval module is to filter out miscellaneous and irrelevant information and find the most relevant context that matches the question. Using the HotpotQA dataset as an example, we employ BERT as the context encoder and use the “[CLS]” token output from BERT as a summary vector for each question/document pair. For each document, we generate an input to feed through BERT by concatenating “[CLS]” + question + “[SEP]” + document + “[SEP]”. Subsequently, we project the output of the “[CLS]” token to compute the relevance score for each document. We label all documents containing supporting facts as relevant. A straightforward approach for gold document classification is to project this summary vector into a single dimension and calculate the binary cross-entropy loss:

$$L = - \sum_{i=0}^n t_i \log P(D_i) + (1 - t_i) \log (1 - P(D_i)),$$

where t_i is the label of D_i , n is the number of documents, and $P(D_i)$ is the probability of document i being in label t_i .

Pre-trained models often perform poorly when dealing with long contexts. Therefore, when processing larger datasets (such as SQuAD), we use a TF-IDF-based Beam Search for candidate paragraph retrieval. To better distinguish between candidate documents, we add an [UNK] token at the beginning of each document. Compared to model-based methods, this approach is more efficient in retrieval and requires fewer computational resources,

while TF-IDF helps rank paragraphs within the documents and align the final results.

3.3.2 Reading comprehension module

At this stage, we implement a multi-task model to extract answers and supporting facts for multi-hop questions, as shown in Fig. 2. First, we obtain candidate documents filtered and ranked by the retrieval module. Then, the context and question are concatenated into a sequence and processed using a Pre-trained Language Model (PLM) called ELECTRA [34] to obtain contextual representations of the input text. Subsequently, we use cross-attention to enhance interactions between the context and the question, achieving better contextual embeddings. Finally, a multi-task learning approach is adopted for answer prediction, where answers and supporting sentences are jointly extracted by simply adding linear prediction layers.

In CausalBridgeQA, we focus on complex questions, specifically those where the model performs poorly on QA data. We propose a “knowledge compensation” mechanism. First, after multiple iterations, we identify incorrect data by comparing it against the standard answers. Then, we select this data, replace the original question q with a question Q generated by a generative model, and integrate the causal relationships extracted during the CRE phase. This enhanced data is fed into the PLM of the reading comprehension module for interpretation. We also incorporate a self-attention mechanism to improve the model’s ability to answer questions effectively. Subsequently, the answer predictor outputs the results. Finally, we conduct multiple iterations, during which the CGC in the CRE phase dynamically adjusts and aggregates intra-sentence and inter-sentence relationships to continuously optimize the second-stage reading comprehension module. The entire process is dynamic and includes a “rollback” mechanism, iterating until a stable final answer is achieved. We describe this process as follows:

The causal relationship R extracted during the CRE stage is represented as:

$$R = \bigcup_{s_i \in S} \bigcup_{p_j \in M(s_i)} \text{Extract}(s_i, p_j) = \{(c_k, e_k)\}_{k=1}^{|R|},$$

where c_k is the Cause of the k -th causal relationship, e_k is the effect of the k -th causal relationship, $M(s_i)$ is the set of causal patterns matching s_i , $\text{Extract}(s_i, p_j)$ extracts the causal pair from sentence s_i and pattern p_j . To represent R in a form usable by the network, we perform an embedding process on it.

$$H_R = \text{Embed}(R) = \{[\text{Embed}(c_k), \text{Embed}(e_k)]\}_{k=1}^{|R|},$$

where Embed is a shared embedding function used to map causal relationships to a vector space. In the reading comprehension module, the causal relationship H_R is integrated to enhance the context representation H_c . The enhanced context representation is defined as:

$$H'_c = \text{Aggregate}(H_c, H_R).$$

Aggregate refers to the fusion operation between context H'_c and causal embedding H_q . To improve QA performance, we further add

a cross-attention interaction module on the PLM to enhance the interaction between context and query representations.

$$Q = H'_c W^Q, \quad K = H_q W^K, \quad V = H_q W^V.$$

The cross-attention calculation adjusted by causal information is

$$\begin{aligned} &\text{CausalAttention}(Q, K, V, H_R) \\ &= \text{softmax}\left(\frac{QK^T}{\sqrt{d_k}} + \text{CausalMask}(H_R)\right)V. \end{aligned}$$

$\text{CausalMask}(H_R)$ is based on the attention mask of H_R , controlling the influence of causal information. Finally, we use the representations obtained through the cross-attention mechanism to jointly predict the answer and supporting facts. Let the start and end positions of the answer be denoted as (s_a, e_a) , and the loss function is

$$\mathcal{L}_{\text{answer}} = -\frac{1}{N} \sum_{i=1}^N [y_{s_a} \log P(s_a) + y_{e_a} \log P(e_a)].$$

Let the set of supporting sentences be denoted as S , and the loss function is

$$\begin{aligned} \mathcal{L}_{\text{support}} = & -\frac{1}{N} \sum_{i=1}^N \sum_{j=1}^M [y_{s_j} \log P(s_j) \\ & + (1 - y_{s_j}) \log (1 - P(s_j))]. \end{aligned}$$

The final loss function of the joint optimization objective is

$$\mathcal{L} = \alpha \mathcal{L}_{\text{answer}} + \beta \mathcal{L}_{\text{support}},$$

where α and β are weight hyperparameters. Up to this point, we fully demonstrate how the causal relationships in the CRE stage are applied to the reading comprehension module.

4 Experiments

In this section, we conduct a series of experiments to demonstrate the effectiveness of our proposed CausalBridgeQA method, along with ablation study and overall evaluation. In summary, we design experiments to address the following research questions:

- **RQ1** Is the causal relationship we have extracted effective? Does it have advantages compared to other current methods for extracting causal relationships? Is the generated question Q semantically richer and of better quality compared to the original multi-hop QA question q ?
- **RQ2** During the MHQA reasoning stage, does our method have an advantage in performance compared to other MHQA tasks, and is there an improvement in model robustness and reasoning ability?
- **RQ3** Does the overall framework we proposed have an accuracy advantage compared to current state-of-the-art methods? Does it solve real-world problems?

4.1 Experimental settings

4.1.1 Datasets

We select three widely recognized and commonly used datasets for

the experiment: HotpotQA [4], SQuAD [35], and TQA [36]. HotpotQA is a dataset specifically designed for testing and evaluating multi-hop QA tasks, and it is also the first dataset to consider model interpretability. It has been widely used in the academic community. SQuAD contains over 100,000 questions from more than 500 Wikipedia articles, and we only use the standard data for training the QA system. TQA includes 26,260 questions from textbooks in life sciences, earth sciences, and physical sciences. We choose this dataset because it contains educational textbook texts, which often express causal relationships.

4.1.2 Evaluation metrics

We find that the causal components in causal relationships are mainly located within the same sentence or adjacent sentences. Therefore, we measure the quality of causal relationship extraction within and between sentences using token-level accuracy, recall, and F1 score. Additionally, we jointly evaluate the causal relationships within the entire input document context, focusing on this as the criterion for assessing the quality of our final extracted causal relationships. For the generation of Q in the CRE phase, we use the ROUGE-L metric to evaluate the quality of the final generated Q by calculating the longest common subsequence between multiple context queries q and the final generated Q . In the MHQA task, we follow the methods used in most previous works, utilizing Exact Match (EM) and F1 score (F1) to evaluate the performance of our method in terms of document retrieval, supporting sentence prediction, and answer extraction.

4.1.3 Baselines and implementation details

We use currently well-performing PLMs as encoders for the MHQA phase to build the overall model: ELECTRA [34] is applied to the HotpotQA dataset, and BERT-large [30] is applied to the SQuAD and TQA datasets, both using the official hyperparameters and then fine-tuned on our task. In the CRE phase, we use a sliding window to examine all possible sentence combinations that may contain causal relationships. We also employ a warm-up strategy in our code implementation, with a warm-up rate of 0.1. After the model has been trained stably over multiple epochs, we identify complex issues and output logs, which is achieved through a rollback mechanism.

We use the Adam optimizer [37] to update the parameters, and all baselines have been adjusted multiple times to find the best hyperparameters. For the TQA dataset, we only use the text data provided by the official source, and all training and evaluation are conducted on a single server equipped with a NVIDIA Tesla A100 GPU.

In the CRE stage, when evaluating the quality of causal relation extraction, we follow the suggestions of CausalQA [14] and CauseNet [15] and include some baseline models for comparison: BERT [30], RoBERTa [38], and UnifiedQA [39]. Since we extract relations based on logical rules and the data has been preprocessed, for a fair comparison, we also compare with some rule-based extraction methods: LogicGuidedQA [10] and CausalText [40]. We also compare with a recently released method based on Structural Causal Models (SCM) [41]. When evaluating the overall performance of the MHQA task, we select the HotpotQA dataset for the main experiments, as it is currently recognized as the best test dataset. For complex problems, we mainly test the effect of multi-hop QA after relation extraction, as detailed in the ablation study. The experimental methods are introduced in Related Work. It is worth noting that we set the evaluation metrics for the HotpotQA dataset as answer span prediction and supporting fact prediction. The model is evaluated based on the EM and F1 for both tasks. For the SQuAD dataset, we use the Dev and Test sets for evaluation, and for the TQA dataset, we only report results on the validation and test sets for text-based questions (true/false and multiple-choice), with accuracy (%) as the metric, all evaluated according to the official standards.

4.2 Main experimental results

4.2.1 Overall performance comparison

RQ1 As shown in Table 1, Intra+Inter represents the overall extraction of causal relationships. We mainly observe the F1 scores across various aspects of the dataset to evaluate the overall performance of the model, with the values highlighted in bold. We observe that CausalBridgeQA achieves significant improvements in multiple areas, surpassing all previously proposed methods. Additionally, we notice that the Intra-sentence test results are always

Table 1 The effectiveness of causal extraction by different method models on the HotpotQA dataset (with incorrect extractions already excluded) tested in terms of extraction accuracy for both intra-sentence and inter-sentence, as well as for full-context input

Model	Intra-sentence			Inter-sentence			Intra+Inter		
	P/%	R/%	F1/%	P/%	R/%	F1/%	P/%	R/%	F1/%
BERT [30]	42.4	32.9	36.8	39.4	30.7	33.4	41.7	31.5	35.4
RoBERTa [38]	46.7	35.2	37.0	44.3	33.5	32.1	45.3	33.7	36.4
UnifiedQA [39]	49.0	38.5	38.1	46.2	36.5	37.7	47.6	36.8	37.1
LogicGuidedQA [10]	47.8	36.6	37.5	45.1	33.4	36.5	46.5	35.3	36.9
CausalText [40]	50.2	39.6	39.4	48.6	39.3	38.6	49.2	38.7	39.1
KPCE [41]	55.3	41.8	40.2	52.8	42.5	41.6	51.3	41.5	40.7
CausalBridgeQA (our)	60.7	45.6	44.8	55.1	44.6	43.8	57.9	47.8	46.0

higher than the Inter-sentence and overall evaluations. This is because extracting causal relationships within a single paragraph or context is much easier and more accurate compared to cross-paragraph extraction. The evaluation metrics reported in Table 1 show values between 35% and 50%, with the highest accuracy reaching only 60%. As mentioned in the introduction, extracting causal relationships in QA datasets is not an easy task. We have kept the examples of extraction errors (those that do not satisfy logical rules) in a text file. Currently, the most suitable task for causal extraction is Event Causality Identification (ECI), and the poor performance on QA datasets is largely related to the type of the data. For the generated question Q , we only measure a ROUGE-L score of around 0.58, which we consider valid (greater than 50%). ROUGE-L tests the longest common subsequence between Q and q . In 58% of the test cases, 87% of the sentences contained key information such as subject and object entities from q . Since the length of different questions q varies, as long as the generated Q includes the key information, it is considered valid.

RQ2 As shown in Table 2, we present a performance comparison of different models on the HotpotQA dataset under interference

Table 2 The test results of CausalBridgeQA on the HotpotQA, SQuAD, and TQA datasets, with most results from the official leaderboard and the original papers

Method	Answer		Supporting	
	EM/%	F1/%	EM/%	F1/%
HotpotQA				
Baseline Model [4]	44.44	58.28	21.95	66.66
QFE [25]	53.70	68.70	58.80	84.70
SAE [6]	66.92	79.62	61.53	86.86
HGN [11]	69.22	82.19	62.76	88.47
DFGN [5]	55.66	69.34	53.10	82.24
AMGN+ [27]	70.53	83.37	63.57	88.83
Transformer-XH [8]	48.95	60.75	41.66	70.01
FE2H on ELECTRA [26]	69.54	82.69	64.78	88.71
CausalBridgeQA (our)	70.61	83.69	65.08	88.86
SQuAD				
BERT [30]	—	—	82.1	84.8
XLNet [42]	86.1	88.8	86.4	89.1
ALBERT [43]	87.4	90.2	88.1	90.9
CausalBridgeQA (our)	88.4	90.8	91.3	91.6
TQA				
XTQA [44]	—	—	41.32	41.67
WSTQ [45]	—	—	64.87	65.15
RAFR [46]	—	—	43.35	41.03
CausalBridgeQA (our)	—	—	71.33	72.64

settings. The values measured in the table are based on the CRE stage processing, incorporating causal knowledge into the training process. Our method shows an absolute improvement of over 40% in both EM and F1 compared to the baseline models. Compared to graph-based models such as HGN and DFGN, we outperform them by more than 10%, achieving a gain of 14.95 in F1 and 14.35 in EM. Our method also demonstrates advantages over the state-of-the-art FE2H and AMGN models. On the SQuAD dataset, we only used some basic models for testing, as most of the questions in SQuAD are single-hop QA with large context spans. Testing on this dataset enhances the model's generalizability and trains the model's ability to extract information from long texts. On the TQA dataset, similar to the HotpotQA dataset, we provide 2–8 answer options for each question and classify the questions as Non-Diagram True/False (NDTF) for interference testing. We focus on factual questions, which are answered using simple facts or named entities. In comparison with the explainable XTQA, we extract span-level explanations to answer the questions. On both the validation and test sets, the text-based questions show a 30% improvement. Our method also outperforms ensemble learning approaches like WSTQ and graph-based methods like RAFR, demonstrating its effectiveness.

4.2.2 Ablation study

To explore the impact of each component in the model on the final results, we conduct ablation experiments, with the results shown in Table 3. PLM refers to the pre-trained language model, and it is considered a key module because the results of causal relationship extraction, the generation of Q , and the attention mechanism are all input here and directly reflected in the outcomes, as described in Subsection 3.3.2. We divide the discussion into two stages: the CRE stage, which involves the construction of the CGC causal entity graph and the generation of Q , and the MHQA stage, which focuses on whether the CRE input is received and the attention mechanism is utilized.

We find that when we remove the causal extraction process and directly tested the MHQA performance, the results are very poor, with a Joint F1 score of 67.18 (corresponding to w/o Pattern Based CGC). After adding Pattern Based CGC, the performance improves. Moreover, when we do not reformulate the question Q , and only add causal relationships as a supplement for complex questions (corresponding to w/o Question Reformulation), the accuracy was 68.18, while the original accuracy is 70.61. Similarly, in the MHQA stage, the self-attention module plays an important role in context modeling, especially for complex questions. We remove the QA

Table 3 Ablation study results on HotpotQA dev set

Setting	PLM	Joint F1/%	Joint EM/%
CRE Stage	w/ Pattern Based CGC	69.88	43.62
	w/o Pattern Based CGC	67.18	42.16
	w/o Question Reformulation	68.18	42.62
MHQA Stage	w/Attention Block	71.28	45.59
	w/o Attention Block	70.36	44.35

model's self-attention module, and regardless of whether one or two cross-attention layers are deleted, both the Joint F1 score and Joint EM decrease. As more attention heads are removed, the model even shows degradation. We provide the following intuitive explanation for this degradation. Since the feedforward network only takes context embeddings as input, disabling the interaction between the question and context may lead to performance degradation. In conclusion, by observing the results in Tables 2 and 3, we can conclude that the knowledge compensation mechanism we design is effective.

4.2.3 Result analysis

RQ3 From the perspective of solving real-world problems, we are more concerned with the quality of predictions for complex questions. Therefore, we conduct experiments after the model training stabilizes: we compare the model's predictions on complex questions with and without the inclusion of Q and the extracted causal relationships. We find that 63% of the questions can be resolved with knowledge injection, 28% remain unanswered (consistent with other models), and 9% lack causal relationships and can only be answered through reasoning. We present a specific test case of CausalBridgeQA on the SQuAD dataset, as shown in Table 4. We use blue to mark causes and red for effects or results. The key causal information in both question q and Q is also annotated to help quickly identify the core elements of the causal relationships. In the MHQA stage, we not only obtain the correct result but also trace the entire reasoning chain for the question. The "Wrong Prediction" refers to the incorrect result obtained by using only q without causal

Table 4 Case study examples on SQuAD datasets

CRE
Context: Because deeper rock is often older, as noted by the principle of superposition, this can result in older rocks moving on top of younger ones. Movement along faults can result in folding, either because the faults are not planar or because rock layers are dragged along, forming drag folds as slip occurs along the fault. Deeper in the Earth, rocks behave plastically, and fold instead of faulting. Cause: Forming drag folds Effect: As slip occurs along the fault Original question q: What happens to rock layers as slip occurs along the fault? Generated Question Q: How does fault slip, influenced by the principle of superposition, lead to the formation of drag folds in rock layers?
MHQA
Correct Answer (q): Forming drag folds. Wrong Prediction (q): Older rocks move below younger rocks. The fault zone becomes wider. The layers fold but do not form drag folds. The rock layers break into smaller pieces. Add Q and causality: How does fault slip, influenced by the principle of superposition, lead to... Full CausalBridgeQA causal chain: Fault slip → Movement of rock layers → Principle of superposition → Older rocks displaced above younger rocks → Dragging of rock layers → Formation of folds → Plastic behavior of rocks → Formation of drag folds.

knowledge, and these errors are mainly due to the model's feature spurious correlations and degradation issues. With multiple incorrect answers, after injecting causal knowledge, CausalBridgeQA not only provides the correct answer to the question but also outputs the complete causal chain of the reasoning process. Overall, our method is effective and capable of solving problems in real-world scenarios.

4.3 More visualization results

Figure 3 shows the overall performance of CausalBridgeQA before and after injecting causal knowledge (CRE stage) across three datasets. Here, we only test the question answering accuracy on the three datasets. In Fig. 3, after adding the extracted causal knowledge (with CRE), the model's question answering accuracy improves across all three datasets, with the largest improvement on the TQA dataset and the smallest on the SQuAD dataset. This is because the original accuracy of the SQuAD dataset in the MHQA task is already quite high, so the improvement here is not very noticeable. Figure 4 represents the results from our targeted testing on the complex question test set, where the evaluation metric is still question answering accuracy. We select complex questions that the initial model could not answer from the three datasets, convert q into Q , and inject causal knowledge. We observe the accuracy of answers to complex questions, with the largest improvement still on the TQA dataset. Overall, accuracy has improved across all the three datasets, indicating that our method can, to some extent, address the false correlations and degradation issues described in the introduction and reduce the model's chances of getting stuck.

In the Causal Relationship Extraction phase, our data is preprocessed. Figure 5 present the statistics for the HotpotQA dataset. As described in Subsection 3.2.1, we follow the

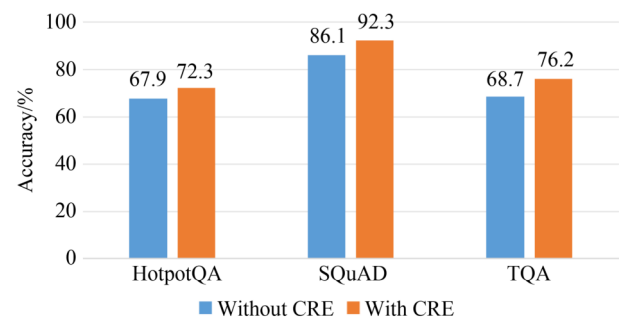


Fig. 3 Overall accuracy of CausalBridgeQA

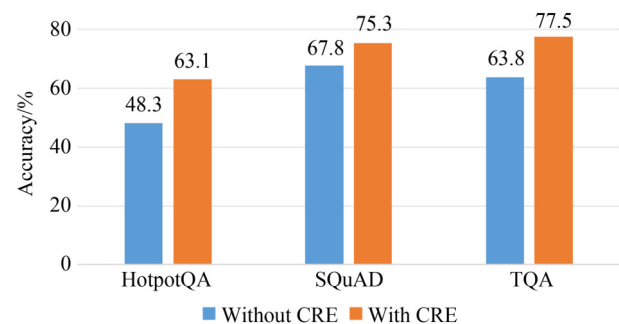


Fig. 4 CausalBridgeQA accuracy on complex questions

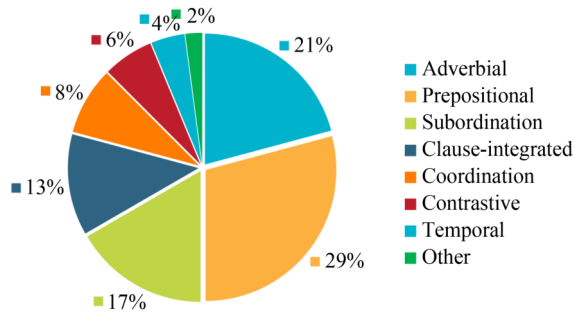


Fig. 5 Causal type statistics

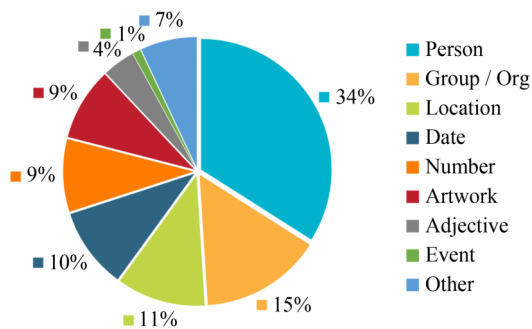


Fig. 6 Question types in HotpotQA

recommendations of [14] and [31], categorizing the causal types in the HotpotQA dataset according to causal typology, which results in seven categories, with other representing non-causal relationships. We incorporate these causal patterns into our evaluation framework to allow improvements in model design. Similarly, following the approach of the original authors of the HotpotQA dataset [4], we remove YES and NO answers and then conduct statistics on the types of questions covered in the HotpotQA dataset. The question types are heuristically extracted, starting from the question words or the prepositions preceding them, as shown in Fig. 6.

5 Conclusion

In this paper, we proposed a novel method called CausalBridgeQA, which primarily addresses a series of issues encountered in current multi-hop question answering reasoning (MHQA) tasks, such as feature spurious correlations, reasoning chain breaks, and model degradation. By extracting causal knowledge and transforming the question format, we incorporated the implicit causal relationships in the original input into the training process, thereby improving the performance of MHQA reasoning tasks. We conducted extensive experiments on three real-world, effective datasets, demonstrating the effectiveness and superiority of the proposed method. We hope that our CausalBridgeQA work will contribute to enhancing the interpretability of future NLP models and promote research in automatic causal relationship extraction.

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Competing interests

The authors declare that they have no competing interests or financial conflicts to disclose.

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